

PURBANCHAL UNIVERSITY



DEPARTMENT OF COMPUTER ENGINEERING KHWOPA ENGINEERING COLLEGE LIBALI-08, BHAKTAPUR

A MID-TERM REPORT ON **Generative AI Based Behavioural Interview Preparation And Guidance System In IT Sector**

Project mid-defence report submitted in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering (Seventh Semester)

Submitted by:

Kriti Koju	(790317)
Malika Budhathoki	(790319)
Rushal Manandhar	(790333)
Xenium Suwal	(790348)

Under the Supervision of
Er. Milan Chikanbanjar

Khwopa Engineering College
Libali-08, Bhaktapur
September, 2026

Acknowledgement

We extend our sincere appreciation to all the individuals who played a pivotal role in the conceptualization and execution of this project. Our profound gratitude goes to Purbanchal University Khwopa Engineering College and the Department of Computer Engineering (HOD. Er. Bikash Chawal and all the faculty members) for providing us with invaluable opportunities to embark on this remarkable endeavour.

Kriti Koju	(790317)
Malika Budhathoki	(790319)
Rushal Manandhar	(790333)
Xenium Suwal	(790348)

Abstract

Interview preparation is an important step for students and job seekers seeking employment opportunities. However, many candidates face challenges with behavioural interview questions, limited access to professional guidance, and insufficient personalized feedback. In addition, individuals are often uncertain about suitable career paths and the skills required for their desired job roles.

This project proposes an Generative AI Based Behavioural Interview Preparation and Guidance System in IT Sector that helps users improve their interview performance and make informed career decisions. The system collects user information through an initial questionnaire regarding career goals, interests, and level to create a personalized profile. Based on this profile, users receive interview questions and learn and practice through an AI chat assistant.

The platform provides both practice and mock interview modes, allowing users to participate in text-based interview sessions. Behavioural interview questions are generated according to the user's selected domain and its level. The system evaluates responses using Artificial Intelligence techniques and provides feedback on performance, helping users identify strengths and areas for improvement.

The proposed system aims to enhance interview readiness for tricky behavioural questions, confidence development, interview simulation, and automated evaluation within a single platform. The system can be beneficial for students, graduates, and job seekers preparing for academic, internship, and professional interviews.

Keywords: *Artificial Intelligence, Interview Preparation, Mock Interview, Natural Language Processing, Performance Evaluation.*

Contents

Acknowledgement	i
Abstract	ii
Contents	v
List of Figures	vi
List of Tables	vii
List of Abbreviations	viii
1 Introduction	1
1.1 Background	1
1.2 Problem Definition	1
1.3 Motivation	2
1.4 Objective	2
1.5 Scope and Application	2
2 Literature Review	3
2.1 Evolution of AI-Based Interview Preparation Systems	3
2.2 Adaptive Question Generation Technique	3
2.3 Semantic Embedding Models for Career Guidance	3
2.4 Retrieval-Augmented Generation for Personalized Guidance	4
2.5 Multi-Dimensional Scoring and Feedback Mechanismss	4
2.6 Existing Challenges in AI Interview and Guidance Systems	5
2.7 Synthesis and Research Gaps	5
3 Project Management	6
3.1 Team Members	6
3.2 Work Breakdown Structure	6
3.3 Software and Hardware Environment	6
3.4 Functional Requirements	7
3.5 Non-Functional Requirements (Validation)	7
4 Feasibility Study	8
4.1 Technical Feasibility	8
4.2 Operational Feasibility	8
4.2.1 Profile Setup and AI Chat Guidance	8
4.2.2 Mock Interview Simulation and Evaluation	8
4.3 Scheduling Feasibility	9
5 Methodology	10
5.1 Overview	10
5.2 Technology Stack	10
5.2.1 Ollama and Llama 3.2 for Question and Answer Generation	10
5.2.2 Groq API for Chat and User Guidance	10
5.2.3 Nomic-Embed-Text for Response Evaluation	11
5.3 System Architecture and Data Flow	11

5.3.1	Block Diagram	11
5.3.2	Context Diagram (DFD Level 0)	12
5.3.3	Level 1 Data Flow Diagram	13
5.4	Use Case Diagram	14
5.5	Sequential Diagram	15
5.6	System Evaluation	16
5.6.1	Evaluation Metrics	16
5.6.2	Retrieval Evaluation	16
5.6.2.1	Similarity Score	16
5.6.2.2	Precision@5	17
5.6.2.3	Recall@5	17
5.6.2.4	Mean Reciprocal Rank	17
5.6.2.5	Retrieval F1	18
5.6.3	Retrieval Evaluation Results	18
5.6.4	Generation Evaluation	19
5.6.4.1	Exact Match	19
5.6.4.2	BLEU	19
5.6.4.3	ROUGE-1	20
5.6.4.4	ROUGE-2	20
5.6.4.5	ROUGE-L	21
5.6.4.6	Token-level F1	21
5.6.5	Generation Evaluation Results	21
5.6.6	Analysis of Generation Results	22
6	Dataset Collection and Analysis	23
6.1	Data Collection and Data Description	23
6.1.1	Data Sources	23
6.1.1.1	Easy-Level Questions	23
6.1.1.2	Medium-Level Questions	23
6.1.1.3	Hard-Level Questions	24
6.1.2	Data Format and Structure	24
6.1.3	Dataset Summary	24
6.2	Data Preprocessing	25
6.2.1	Processed Dataset Summary	25
6.3	Exploratory Data Analysis (EDA)	26
6.3.1	Easy-Level Dataset Analysis	26
6.3.2	Medium-Level Dataset Analysis	27
6.3.3	Hard-Level Dataset Analysis	28
6.3.4	Comparative Analysis Across All Difficulty Levels	29
6.3.5	Dataset Coverage Summary	30
7	Results And Discussion	31
7.1	Work Done	31
7.1.1	System Development and Interview Question Dataset	31
7.1.2	Question Retrieval Process	31
7.1.3	Retrieval Evaluation and Analysis	31
7.1.4	Context-Based Response Generation	32
7.1.5	Mock Interview Functionality	32

7.2	Web Interface	32
7.2.1	Home Page	32
7.2.2	Account Creation	33
7.2.3	AI Chat	34
7.2.4	Mock Interview	35
7.3	Remaining Work	36
7.3.1	User Profile and Dashboard	36
7.3.2	User Interview Evaluation	36
7.3.3	Interview Score and Feedback Presentation	36
7.3.4	Interview History	36
Bibliography		37

List of Figures

5.1	System Block Diagram and Workflow	11
5.2	DFD Level 0 - System Context	12
5.3	DFD Level 1 - Detailed Data Flow	13
5.4	Use Case Diagram	14
5.5	Sequence Diagram	15
5.6	Reterival Metrics Result	19
5.7	Generation Metrics Result	22
6.1	Bar Chart for Easy Questions per Role	26
6.2	Bar Chart for Medium Questions per Role	27
6.3	Bar Chart for Hard Questions per Role	28
6.4	Comparative Analysis Across All Difficulty Levels	29
7.1	Home Page	32
7.2	Option Page	33
7.3	Account Creation Interface	33
7.4	AI Chat Interface	34
7.5	Mock Interview Interface	35

List of Tables

3.1	Functional Requirements	7
5.1	Retrieval Evaluation Metrics	18
5.2	Generation Evaluation Metrics	22
6.1	Summary of Data Sources by Difficulty Level	24
6.2	Summary of Data Sources by Difficulty Level	25

List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BELU	Bilingual Evaluation Understudy
CSS	Cascading Style Sheets
DB	Database
DFD	Data Flow Diagram
GB	Gigabyte
HR	Human Resources
HTML5	HyperText Markup Language version 5
IoT	Internet of Things
IT	Information Technology
JSON	JavaScript Object Notation
LLM	Large Language Model
MRR	Mean Reciprocal Rank
NLP	Natural Language Processing
QA	Quality Assurance
RAG	Retrieval-Augmented Generation
RAM	Random Access Memory
REST	Representational State Transfer
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
TF-IDF	Term Frequency-Inverse Document Frequency
UX	User Experience
XLSX	Excel Spreadsheet Format

Chapter 1

Introduction

1.1 Background

The recruitment process has become increasingly competitive, with employers evaluating not only academic qualifications and technical skills but also communication, confidence, and problem-solving ability during behavioural interviews. As a result, behavioural interview preparation has become essential for students and job seekers.

However, many candidates lack access to professional guidance, realistic practice environments, and personalized feedback. Traditional methods, such as reading sample questions or watching tutorials, offer only general knowledge and fail to address individual skill gaps like clarity of communication or structured thinking.

Recent research has explored AI-driven solutions to this gap. AI-based guidance platforms have begun combining conversational assistants with personalized recommendations, while dedicated interview-preparation platforms have introduced AI-generated, role-specific questions with multi-dimensional feedback covering technical, communication, and problem-solving skills. These efforts confirm that combining conversational AI, personalized question generation, and structured evaluation meaningfully improves behavioural interview readiness compared to static preparation methods.

Building on these directions, this project proposes a Generative AI-Based Behavioural Interview Preparation and Guidance System that allows users to manage their profile, engage in conversational chat with an AI assistant for queries and guidance, participate in AI-generated mock behavioural interview sessions, and receive automated evaluation, scores, feedback, and progress reports. The system also includes an administrative module through which users, behavioural interview questions, and system activity can be managed, along with reporting and analytics support.

1.2 Problem Definition

Existing AI-based solutions address the behavioural interview-readiness problem only partially. Some guidance platforms use conversational AI to answer general queries, but their focus remains on advisory interaction rather than running structured mock behavioural interviews. Conversely, AI-based interview platforms that generate role-specific questions and score responses across technical, communication, and problem-solving dimensions are largely built for one-time evaluation, with limited support for an ongoing assistant that candidates can consult for general guidance alongside practice.

This leaves a gap: users lack a single platform to chat with an AI assistant for guidance, manage a profile of their skills and interests, practice through adaptive mock behavioural interviews, and view a consolidated record of their scores, feedback, and progress. Many

existing systems also lack sufficient personalization and explainable feedback, with unclear scoring and potential bias reducing user trust. This motivates an integrated AI-powered system combining profile management, chat-based guidance, mock behavioural interview simulation, adaptive questioning, and NLP-based evaluation with transparent feedback and reporting.

1.3 Motivation

Rising competition and evolving industry demands make behavioural interview readiness increasingly important. Many candidates with strong academic knowledge still struggle in interviews due to nervousness, incomplete responses, and difficulty presenting their skills effectively. Existing platforms typically offer static question sets with limited, non-personalized feedback. This project is motivated by the need for an intelligent system that combines conversational guidance, mock behavioural interview practice, and performance evaluation into a single platform, delivering personalized, adaptive, and explainable behavioural interview preparation.

1.4 Objective

The main objective of this project is:

- To develop a Generative AI-Based Behavioural Interview Preparation and Guidance System
 - It provides users with conversational AI guidance, personalized mock behavioural interview practice, and automated performance evaluation through explainable scoring, feedback, and progress reports.

1.5 Scope and Application

The proposed system helps students, graduates, and job seekers prepare for behavioural interviews through AI-powered guidance and mock interview practice.

- Users register or log in and build a profile with their skills and interests.
- Users interact with a Local AI/LLM agent through open chat for general guidance.
- Users participate in structured, AI-generated mock behavioural interview sessions.
- Responses are evaluated by the AI agent, generating scores, feedback, and progress reports.
- An administrative interface manages users, behavioural interview questions, and system reports.
- The current implementation is text-based, with a modular design allowing future extension to other interaction modes.

Chapter 2

Literature Review

2.1 Evolution of AI-Based Interview Preparation Systems

Interview preparation has traditionally relied on peer practice, mock sessions with career counselors, or generic online question banks, all of which are inconsistent in quality and limited in accessibility. The introduction of generative AI has enabled a shift toward automated, personalized preparation platforms that can adapt to an individual candidate’s target role rather than offering static, one-size-fits-all content. Kulkarni et al. proposed PrepWise, a generative AI-powered platform that generates interview questions dynamically based on a candidate’s selected job role, allowing the same underlying system to remain relevant across a wide range of job categories without manual curation of question sets [1]. This evolution reflects a broader trend in the field: rather than treating interview preparation as a fixed content-delivery problem, modern systems treat it as a personalized, data-driven pipeline that adapts to each user’s profile, chosen role, and interaction history.

2.2 Adaptive Question Generation Technique

A core capability underpinning both interview preparation and guidance systems is the ability to generate contextually relevant questions rather than drawing from a fixed, pre-written bank. PrepWise generates role-specific interview questions using a large language model accessed through the OpenRouter API, tailoring each question set to the job role selected by the candidate, and dynamically adjusts follow-up questioning based on the candidate’s prior responses during a session [1]. This approach allows the system to simulate a more natural, conversational interview flow rather than presenting a static, sequential list of unrelated questions. Similarly, Gharte et al. generate conversational responses through a locally-deployed Mistral 7B model that adapts its output based on retrieved context relevant to the user’s query, demonstrating that adaptive, LLM-driven generation is a shared architectural pattern across both interview simulation and open-ended guidance use cases [2]. This shared reliance on LLM-driven generation, rather than static content, supports the design choice of using an open-source model such as Llama for both structured interview questioning and free-form guidance conversation within a single system.

2.3 Semantic Embedding Models for Career Guidance

Providing meaningful, personalized career guidance requires the system to understand user queries, career paths, and skill requirements at a semantic level rather than relying

on rigid keyword matching. Gharte et al. adopted the all-MiniLM-L6-v2 sentence embedding model to compute semantic similarity between user profiles, career descriptions, and job postings, storing these embeddings in a ChromaDB vector database for efficient similarity search. The authors report that this embedding model outperformed both traditional TF-IDF vectorization and a multilingual paraphrase-mpnet-based alternative during internal validation, indicating that the choice of embedding model has a measurable impact on the quality of semantic matching and, by extension, the relevance of guidance provided to the user [2]. This finding directly supports the use of a modern sentence-embedding model as the foundation for matching a user’s stated interests or goals to relevant career information within a guidance-oriented chat system.

2.4 Retrieval-Augmented Generation for Personalized Guidance

Generating open-ended career guidance, such as advising a user on what skills to develop or which direction to pursue, requires a mechanism that keeps the AI’s responses grounded in the user’s own context rather than producing generic, ungrounded advice. Gharte et al. implement this through a Retrieval-Augmented Generation (RAG) chatbot: when a user submits a query, the system retrieves the top three most relevant context chunks from its vector database and injects them into the prompt before the language model generates a response, ensuring that guidance remains personalized and grounded in the user’s stored history rather than defaulting to generic language model output [2]. This retrieval-based grounding mechanism is a significant advance over earlier chatbot-style guidance tools, which typically rely on static FAQ matching or ungrounded generation. It is worth noting that this pattern of retrieval-grounded, text-based guidance parallels developments in adjacent multimodal systems; more advanced interview simulators such as SimInterview extend a similar retrieval-based approach into a spoken interaction format, combining automatic speech recognition with speech synthesis so that a candidate can conduct an entire mock interview by voice rather than text, suggesting that a text-first RAG architecture, such as the one reviewed here, could later be extended to support voice-based interaction without requiring a fundamentally different underlying design [3].

2.5 Multi-Dimensional Scoring and Feedback Mechanisms

A recurring theme across both reviewed systems is the move away from single-score evaluation toward multi-dimensional, explainable scoring. PrepWise evaluates candidate responses across three distinct dimensions, technical proficiency, communication skills, and problem-solving ability, and logs each session with its score and feedback into an Interview History dashboard so users can track their performance trend over multiple attempts [1]. Gharte et al. take a comparable but more explicitly formalized approach with a centralized scoring engine that combines Resume Strength, Job Fit, and Career Readiness into a single weighted score, where the weighting coefficients adjust dynamically based on user context, for example, giving recent graduates higher weighting on career readiness and experienced professionals higher weighting on job fit [2]. Notably,

this adaptation occurs entirely at the scoring layer without requiring model retraining, making the mechanism lightweight and easy to extend to new scoring dimensions. An adaptive feedback loop further refines this over time, with users rating recommendations as relevant or irrelevant, which the authors project improves recommendation relevance from approximately 71–72% to approximately 93% [2]. However, in both papers, these evaluation outcomes are self-reported through descriptive walkthroughs or preliminary, design-based testing rather than rigorous, large-scale user studies; PrepWise reports no quantitative accuracy metrics at all [1], while Gharte et al. explicitly characterize their figures as "estimated" and derived from "controlled preliminary testing" rather than a deployed system with real users [2].

2.6 Existing Challenges in AI Interview and Guidance Systems

Although the reviewed systems demonstrate meaningful progress, several challenges remain across the current body of work. Existing platforms often focus on either interview simulation or career guidance individually rather than fully integrating both within a single, cohesive system; PrepWise offers a complete interview preparation pipeline but does not provide any open-ended career-path guidance functionality [1], while the system proposed by Gharte et al. offers strong guidance and recommendation capabilities but does not include a structured mock interview or scoring module [2]. In addition, both systems report limited transparency regarding the rigor of their evaluation: PrepWise provides no quantitative accuracy, precision, or user-study metrics to validate its scoring outputs [1], and the career recommendation system explicitly characterizes its reported accuracy and satisfaction figures as design-based estimates from preliminary, controlled testing rather than outcomes from a system deployed to real users at scale [2]. Concerns around scalability, dependency on external or locally-hosted large language models, and the need for adaptive, context-aware scoring that can generalize across different job roles and career domains also remain open issues. These limitations collectively indicate the need for a more comprehensive, transparently evaluated platform that unifies interview preparation and career guidance within a single system.

2.7 Synthesis and Research Gaps

The reviewed literature demonstrates that generative AI can meaningfully support two related but distinct needs: structured interview simulation with adaptive questioning and multi-dimensional scoring [1], and semantic, retrieval-augmented career guidance grounded in a user's profile and interaction history [2]. However, no existing system integrates a complete, text-based interview preparation pipeline with a retrieval-augmented guidance module offering personalized career-path recommendations within one platform. Both reviewed systems also share a common limitation: neither reports results validated on a deployed system with real users at scale. Therefore, there is a clear need for a Generative AI-Based Behavioural Interview Preparation and Guidance System that combines mock interview simulation, profile tracking, and semantic embedding-based career guidance into a single, text-based platform, supported by a more concrete evaluation methodology than currently reported in comparable systems.

Chapter 3

Project Management

3.1 Team Members

The project will be carried out with the contribution of the following four team members.

1. Kriti Koju (790317)
2. Malika Budhathoki (790319)
3. Rushal Manandhar (790333)
4. Xenium Suwal (790348)

3.2 Work Breakdown Structure

S.N	Job Description	Duration (days)	Week			
			1 st - 3 rd	4 th - 6 th	7 th - 9 th	10 th - 12 th
1.	Problem Identifica- tion	6				
2.	Analysis	11				
3.	Design	14				
4.	Coding	28				
5.	Documentation	30				

3.3 Software and Hardware Environment

The development environment for the Generative AI-Based Interview Preparation and Guidance System has been configured using modern web development and Generative AI technologies. The system operates with the following software and hardware specifications:

- **Back-end & AI:** Python, Django/Django REST Framework for backend development and Generative AI APIs/models.
- **Front-end:** React.js, HTML5, Tailwind CSS, and JavaScript for developing the interactive user interface.
- **Database:** PostgreSQL for storing user accounts, interview-related data, questions, answers, scores, and feedback.
- **Development Tools:** Visual Studio Code, Git/GitHub, Docker and Docker Compose for development, version control, and containerized deployment.

- **Hardware:** The system has been developed and tested on a standard computer with at least **8 GB RAM**, a modern multi-core processor, and sufficient storage for datasets, dependencies, and application containers.

3.4 Functional Requirements

Table 3.1 lists the functional requirements of the system along with their current implementation status.

Table 3.1: Functional Requirements

SN	Requirement	Status
01	User Registration and Login	Completed
02	Select IT Job Role to build profile	In Progress
03	Conduct Mock Interview and Submit Answers	Completed
04	AI-Based Answer Evaluation and Scoring	In Progress
05	Chat with AI and get Guidance	Completed

3.5 Non-Functional Requirements (Validation)

During the implementation phase, the following non-functional attributes are considered:

- **Reliability:** The system should consistently generate relevant questions and provide appropriate evaluation and feedback.
- **Performance:** The system should respond to user requests within a reasonable time.
- **Security:** User accounts and interview data should be protected through secure authentication and authorization.
- **Maintainability:** The system should be modular and easy to update, debug, and extend.
- **Usability:** The interface should be simple and easy for users to navigate during interview preparation.

Chapter 4

Feasibility Study

4.1 Technical Feasibility

The proposed Generative AI-Based Behavioural Interview Preparation and Guidance System utilizes Large Language Models (LLMs), sentence embedding models for semantic retrieval, and NLP-based evaluation techniques. Core functionalities such as AI-generated mock interview questions, conversational guidance, and automated response evaluation are implemented using open-source AI models served through Ollama and the Groq API.

The backend is built with Python and Django REST Framework, the frontend with React.js, HTML5, Tailwind CSS, and JavaScript, and PostgreSQL is used for storing user profiles, interview sessions, and results. Text embeddings for semantic question retrieval are generated using the *nomic-embed-text* model via Ollama, enabling context-aware question generation without model retraining.

Since all required technologies are mature, well-documented, and require only standard computing resources, the proposed system is technically feasible.

4.2 Operational Feasibility

The system is designed to be simple and user-friendly for students, graduates, and job seekers preparing for behavioural interviews. Users register an account, log in, and build their profile by selecting their skills, interests, and target IT job role. Based on this profile, the user can either chat with the AI assistant for general guidance or participate in a structured mock interview session and receive automated evaluation and feedback.

The main parts of the system are:

4.2.1 Profile Setup and AI Chat Guidance

The system collects the user's skills, interests, and selected job role during profile setup, which is stored in the database. Users can engage in open conversation with an AI assistant at any time to ask interview-related questions, request tips, or seek general career guidance. This conversational layer uses a large language model to generate context-aware, profile-grounded responses.

4.2.2 Mock Interview Simulation and Evaluation

The mock interview module generates personalized behavioural interview questions based on the user's selected role and, for returning users, their past practice history.

Users respond to each question through text. The system evaluates responses using semantic similarity (cosine similarity between the user's answer and reference answer embeddings) and provides scores, feedback, and progress reports. Integration of the profile, chat, and mock interview modules allows users to prepare for interviews in a realistic, adaptive, and interactive environment.

With all modules integrated into a single platform, the proposed system can effectively assist users in preparing for behavioural interviews and tracking their improvement over time.

4.3 Scheduling Feasibility

The project can be completed within the allocated project duration through proper planning and task management. Activities such as requirement analysis, system design (block diagram, DFDs, use case diagram), implementation of the profile, chat, mock interview, and evaluation modules, testing, and documentation will be performed according to the project schedule. Regular progress monitoring, periodic reviews, and documentation updates will ensure that project milestones are achieved on time. Therefore, the proposed system is scheduling feasible and can be completed within the specified timeframe.

Chapter 5

Methodology

5.1 Overview

This chapter presents the methodology adopted for developing the Generative AI-Based Behavioural Interview Preparation and Guidance System. The system combines locally deployed language models with cloud-based AI services to generate personalized interview questions, evaluate candidate responses, and provide real-time guidance. The architecture integrates three core AI technologies: Ollama for local model deployment, Llama 3.2 for question and answer generation, Groq API for conversational chat and guidance, and nomic-embed-text for semantic similarity-based evaluation.

5.2 Technology Stack

5.2.1 Ollama and Llama 3.2 for Question and Answer Generation

Ollama is an open-source framework that enables local deployment of large language models on standard hardware. In this system, Ollama hosts **Llama 3.2**, Meta's open-source language model, which is used to:

- **Generate behavioural interview questions:** Based on the user's selected IT job role and difficulty level (Easy, Medium, Hard), Llama 3.2 generates contextually appropriate questions. The model receives structured prompts containing the target role, difficulty tier, and past practice history to ensure question variety and relevance.
- **Generate reference answers:** For each generated question, the model produces ideal reference answers that serve as benchmarks for evaluating candidate responses.
- **Local deployment benefits:** Running Llama 3.2 locally via Ollama eliminates recurring API costs, ensures user data privacy (responses remain on-device), and reduces latency during question generation.

5.2.2 Groq API for Chat and User Guidance

Groq is a cloud-based AI inference platform used specifically for the chat and guidance module. When users ask interview-related questions or request career guidance, the system processes these queries through Groq API, which provides:

- **Real-time conversational responses:** Users can ask open-ended questions about interview preparation, skill development, and career paths. Groq handles these requests with ultra-low latency, ensuring responsive chat interactions.

- **Personalized guidance:** The system sends user profile information (skills, interests, selected role) to Groq, enabling context-aware, personalized advice without requiring local model deployment.
- **Scalability:** During high usage periods, Groq’s optimized cloud infrastructure handles concurrent chat requests efficiently, preventing local hardware bottlenecks.
- **Implementation:** When a user submits a chat query, the system constructs a prompt containing the query, user context, and previous conversation history, sends it to Groq API (using Llama 3.2 model), and streams the response back to the user interface.

5.2.3 Nomic-Embed-Text for Response Evaluation

Nomic-embed-text is a sentence embedding model used to evaluate user responses during mock interviews. The evaluation process works as follows:

- **Embedding generation:** During system initialization, all reference answers are converted into 384-dimensional semantic embeddings and stored in the database.
- **Response evaluation:** When a user submits an answer during a mock interview, the system:
 1. Embeds the user’s response using nomic-embed-text
 2. Retrieves the pre-computed embedding of the reference answer
 3. Computes cosine similarity between the two embeddings
- **Similarity scoring:** The cosine similarity score (0 to 1) indicates semantic alignment between the user’s answer and the ideal reference response, capturing paraphrased or contextually equivalent answers that keyword-based matching would miss.
- **Local execution:** Embedding and similarity computation are performed locally via Ollama, keeping evaluation lightweight and cost-free.

5.3 System Architecture and Data Flow

5.3.1 Block Diagram

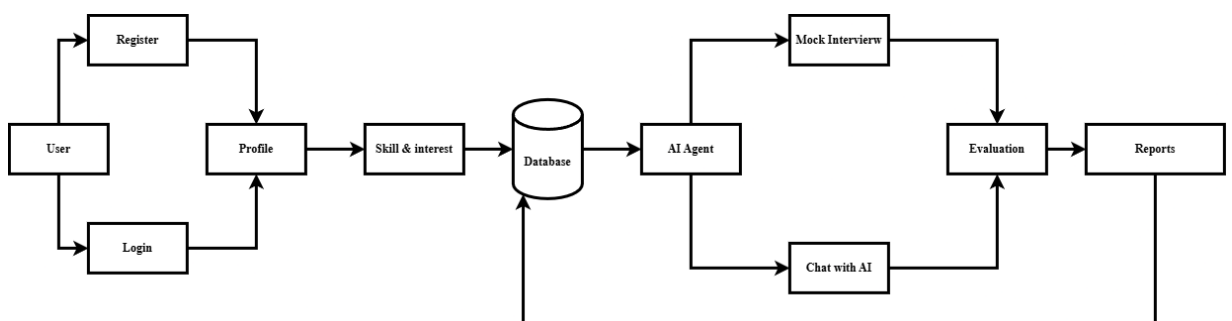


Figure 5.1: System Block Diagram and Workflow

The block diagram represents the overall workflow of the system. Users register and build a profile with their skills and interests stored in the Database. The AI Agent retrieves profile information to generate personalized interview questions via Ollama (Llama 3.2). During mock interviews, user responses are evaluated through semantic embeddings (nomic-embed-text) and scoring. Users can also interact with the chat module for guidance queries, which are processed through Groq API. Evaluation results are used to generate Reports with scores, feedback, and progress tracking.

5.3.2 Context Diagram (DFD Level 0)

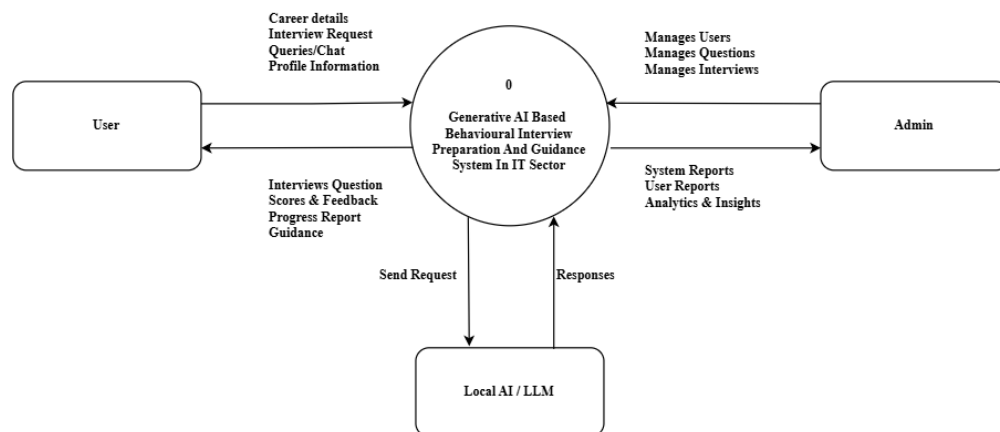


Figure 5.2: DFD Level 0 - System Context

The Context Diagram represents the entire system as a single process interacting with three external entities: User, Admin, and Local AI/LLM. The User sends profile information, interview requests, and chat queries; receives interview questions, scores, feedback, and guidance. The Admin manages users, questions, and interviews; receives system reports and analytics. The Local AI/LLM (Ollama + Llama 3.2 + Groq) acts as the AI engine for question generation, answer generation, and chat guidance.

5.3.3 Level 1 Data Flow Diagram

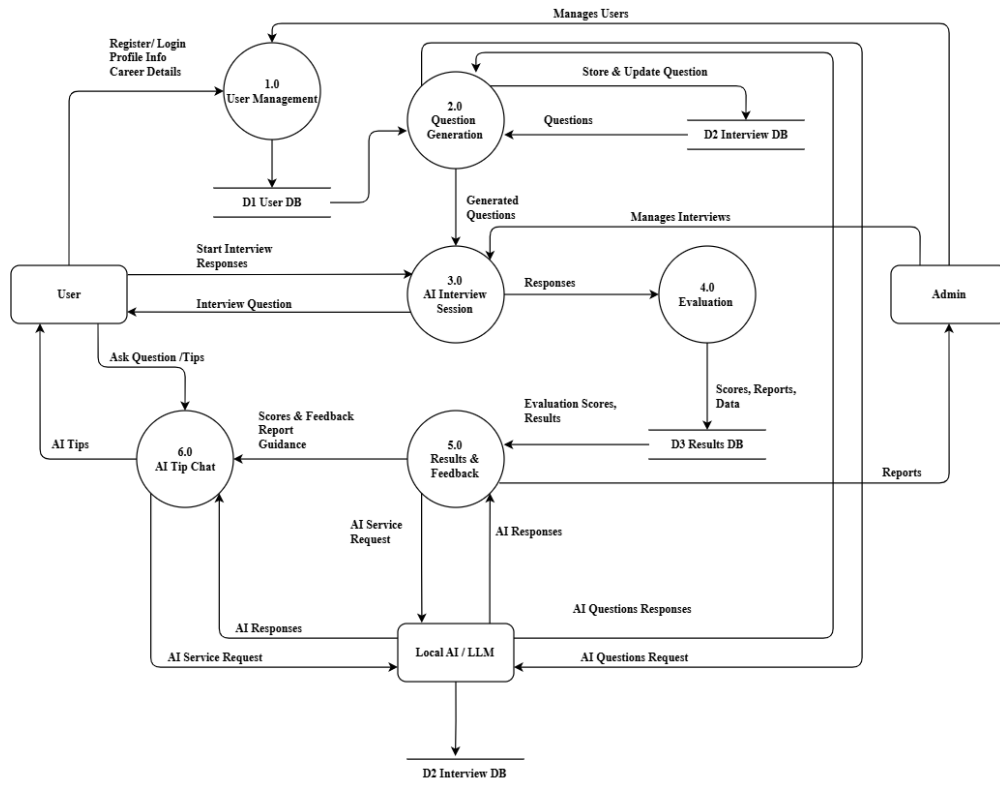


Figure 5.3: DFD Level 1 - Detailed Data Flow

This diagram breaks down the system into six sub-processes and three data stores:

1. **1.0 User Management:** Handles registration/login and profile setup, storing user data in D1 User DB.
2. **2.0 Question Generation:** Retrieves user profile and requests AI-generated questions from Ollama (Llama 3.2). Generated questions are stored in D2 Interview DB.
3. **3.0 AI Interview Session:** User receives questions and submits responses during mock interviews.
4. **4.0 Evaluation:** Scores user responses using nomic-embed-text embeddings and cosine similarity, storing results in D3 Results DB.
5. **5.0 Results and Feedback:** Converts raw scores into feedback and progress reports, calling on Llama 3.2 (via Ollama) for explanatory text.
6. **6.0 AI Chat:** Processes user queries through Groq API for real-time guidance and interview tips.

Throughout the workflow, Ollama serves Llama 3.2 for question/answer generation (processes 2.0, 5.0), Groq API handles chat interactions (process 6.0), and nomic-embed-text performs lightweight embedding and similarity computation (process 4.0).

5.4 Use Case Diagram

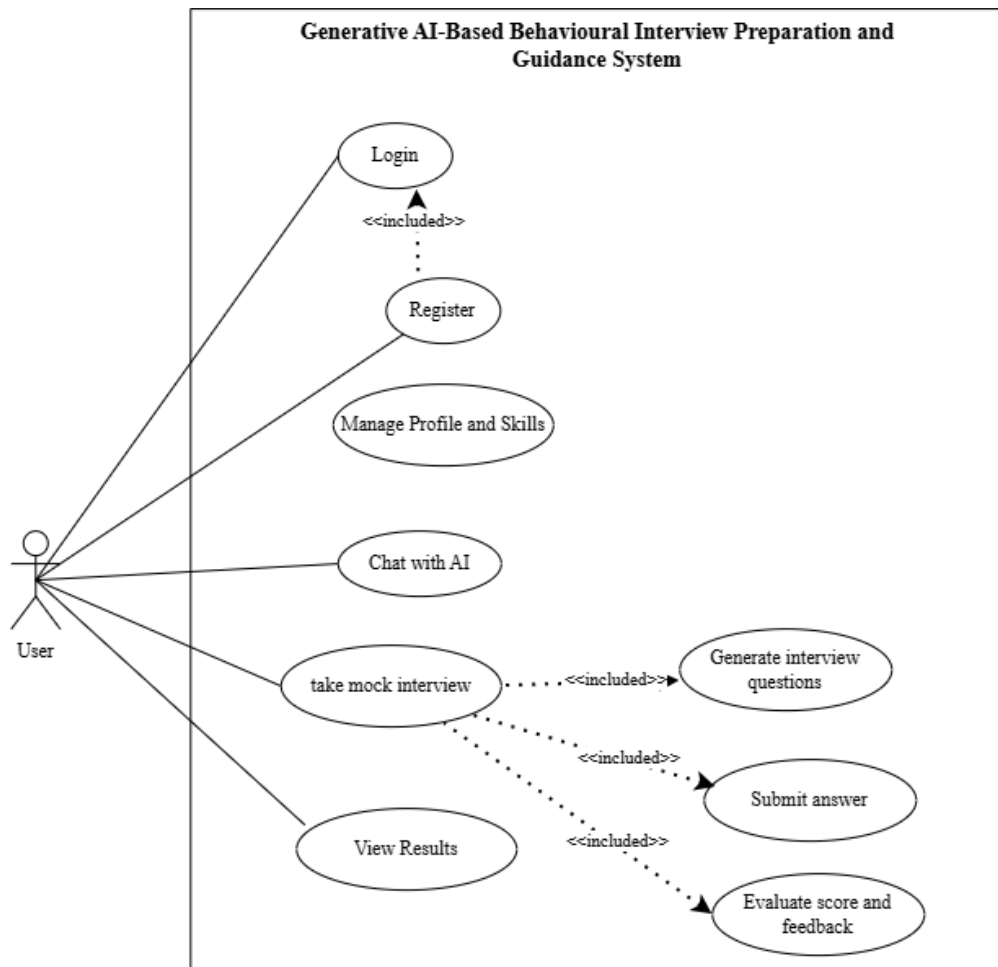


Figure 5.4: Use Case Diagram

The use case diagram represents the main interactions between the user and the Generative AI-Based Behavioural Interview Preparation and Guidance System. The user can register and log in to access the system, manage their profile and skills, and communicate with the AI through the chat feature. The user can also take a mock interview, where the system generates interview questions, accepts the user's answers, and evaluates them to provide scores and feedback. The user can then view the overall interview results. The «included» relationships indicate that generating questions, submitting answers, and evaluating the responses are essential parts of the mock interview process.

5.5 Sequential Diagram

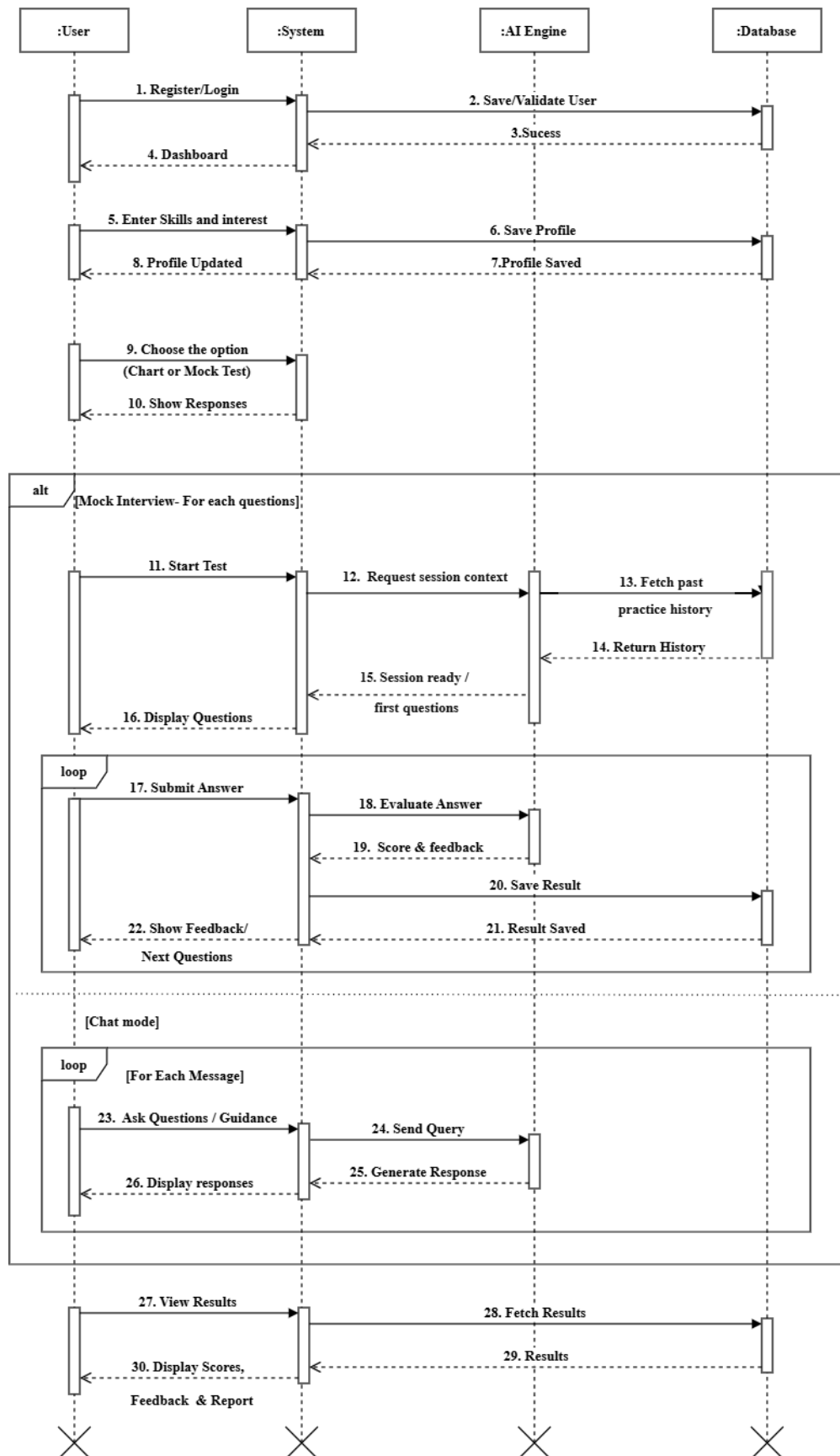


Figure 5.5: Sequence Diagram

The sequence diagram illustrates the interaction flow:

- **Registration and Profile Setup:** User logs in and completes profile with skills and interests.
- **Mock Interview Mode:** System fetches user's past practice history once, then generates context-aware questions using Ollama (Llama 3.2). After each response is submitted, it is evaluated via nomic-embed-text embeddings and cosine similarity, and the next question is generated.
- **Chat Mode:** User asks interview-related questions or requests guidance. Each query is sent to Groq API, which generates personalized responses based on user profile.
- **Results and Reports:** System compiles scores, feedback, and progress reports retrieved from the database.

5.6 System Evaluation

The proposed AI-based interview preparation system was evaluated to assess the performance of its two main components: the question retrieval component and the answer generation component. The evaluation was performed using a set of 40 interview questions containing the expected question, role, keywords, and ideal answer.

The evaluation was divided into two stages. First, the retrieval component was evaluated to determine how effectively the system identifies relevant interview questions from the knowledge base. Second, the generation component was evaluated by comparing the generated responses with the corresponding reference answers.

5.6.1 Evaluation Metrics

A selected set of metrics was used to evaluate the retrieval and generation components. Similarity Score, Precision@5, Recall@5, Mean Reciprocal Rank (MRR), and F1 were used for retrieval evaluation. Exact Match, BLEU, ROUGE-1, ROUGE-2, ROUGE-L, and Token-level F1 were used for evaluating generated responses.

5.6.2 Retrieval Evaluation

5.6.2.1 Similarity Score

The Similarity Score measures the semantic similarity between the user's input question and the questions retrieved from the vector database. In the implemented system, the score is derived from the distance returned by the vector retrieval process as:

$$\text{Similarity Score} = 1 - \text{Distance} \quad (5.1)$$

For each input question, the system retrieves the top five most similar questions. The similarity scores of the retrieved questions are averaged to obtain the final similarity score for that query.

The average similarity score obtained from the evaluation was:

$$\boxed{\text{Similarity Score} = 0.685} \quad (5.2)$$

This indicates that the retrieved questions have a moderate to relatively high semantic similarity to the input questions.

5.6.2.2 Precision@5

Precision@5 measures the proportion of relevant questions among the five questions retrieved by the system. It is calculated as:

$$\text{Precision@5} = \frac{\text{Number of relevant questions in top 5}}{5} \quad (5.3)$$

The system achieved:

$$\boxed{\text{Precision@5} = 0.252} \quad (5.4)$$

This indicates that approximately 25.2% of the retrieved results within the top five positions were considered relevant according to the evaluation criterion.

5.6.2.3 Recall@5

Recall@5 measures how many of the relevant questions available for a query are successfully retrieved within the top five results. It is calculated as:

$$\text{Recall@5} = \frac{\text{Number of relevant questions retrieved in top 5}}{\text{Total number of relevant questions}} \quad (5.5)$$

The system achieved:

$$\boxed{\text{Recall@5} = 0.365} \quad (5.6)$$

This indicates that approximately 36.5% of the relevant questions were retrieved within the top five results.

5.6.2.4 Mean Reciprocal Rank

Mean Reciprocal Rank (MRR) measures how highly the first relevant retrieved question is ranked. It is calculated as:

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i} \quad (5.7)$$

where N is the total number of evaluation queries and rank_i is the position of the first relevant result for query i .

The system achieved:

$$\boxed{\text{MRR} = 0.958} \quad (5.8)$$

The high MRR value indicates that when a relevant result is retrieved, it is generally positioned very close to the top of the retrieval results.

5.6.2.5 Retrieval F1

F1 provides a combined measure of precision and recall. It is calculated as the harmonic mean of precision and recall:

$$F1 = \frac{2 \times P \times R}{P + R} \quad (5.9)$$

where P represents precision and R represents recall.

The system achieved:

$$\boxed{F1 = 0.295} \quad (5.10)$$

This value reflects the combined retrieval performance considering both precision and recall.

5.6.3 Retrieval Evaluation Results

The retrieval performance of the proposed system is summarized in Table 5.1.

Table 5.1: Retrieval Evaluation Metrics

Metric	Average Score
Similarity Score	0.685
Precision@5	0.252
Recall@5	0.365
MRR	0.958
F1	0.295

The retrieval results show that the system obtains a relatively high similarity score of 0.685 and a high MRR of 0.958. The MRR indicates that the first relevant result is generally ranked close to the top. However, the Precision@5 and Recall@5 scores are comparatively lower, with values of 0.252 and 0.365 respectively. The retrieval F1 score of 0.295 reflects the balance between the precision and recall values.

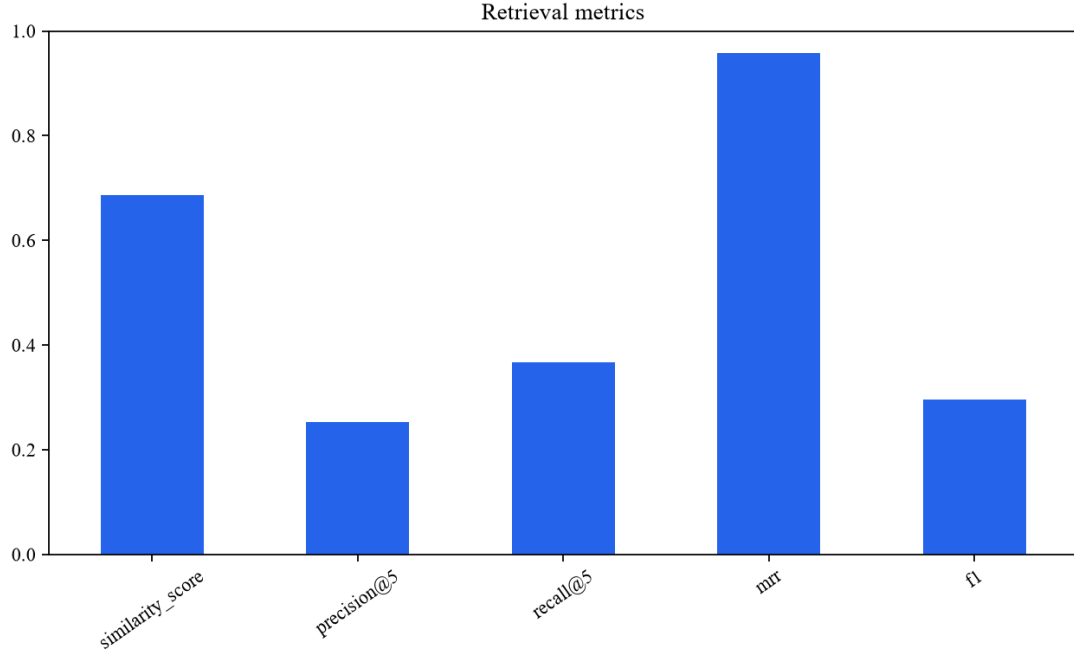


Figure 5.6: Reterival Metrics Result

5.6.4 Generation Evaluation

The answer generation component produces an interview response using the retrieved questions and their associated contextual information. The generated response is compared with the corresponding reference answer using Exact Match, BLEU, ROUGE-1, ROUGE-2, ROUGE-L, and Token-level F1.

5.6.4.1 Exact Match

Exact Match measures whether the generated answer exactly matches the reference answer after the normalization applied by the evaluation system.

It can be expressed as:

$$\text{Exact Match} = \frac{\text{Number of exactly matched responses}}{\text{Total number of responses}} \quad (5.11)$$

The system achieved:

$$\boxed{\text{Exact Match} = 0.100} \quad (5.12)$$

This indicates that approximately 10% of the generated responses achieved an exact match with their corresponding reference responses.

5.6.4.2 BLEU

BLEU measures the overlap between the generated response and the reference answer using matching n-grams, together with a brevity penalty.

The system achieved:

$$\boxed{BLEU = 0.170} \quad (5.13)$$

This indicates a relatively limited n-gram overlap between the generated responses and the reference answers.

5.6.4.3 ROUGE-1

ROUGE-1 measures the overlap of individual words, or unigrams, between the generated response and the reference answer.

The ROUGE-1 F1 score is calculated using the harmonic mean of unigram-level precision and recall:

$$ROUGE-1 = \frac{2 \times P_1 \times R_1}{P_1 + R_1} \quad (5.14)$$

where P_1 and R_1 represent unigram precision and recall, respectively.

The system achieved:

$$\boxed{ROUGE-1 = 0.345} \quad (5.15)$$

This indicates that there is a moderate level of unigram overlap between the generated responses and the reference answers.

5.6.4.4 ROUGE-2

ROUGE-2 measures the overlap of two-word sequences, or bigrams, between the generated response and the reference answer.

The F1-based ROUGE-2 score can be expressed as:

$$ROUGE-2 = \frac{2 \times P_2 \times R_2}{P_2 + R_2} \quad (5.16)$$

where P_2 and R_2 represent bigram precision and recall, respectively.

The system achieved:

$$\boxed{ROUGE-2 = 0.210} \quad (5.17)$$

This indicates that the generated responses share a lower level of consecutive two-word sequence overlap with the reference answers than the unigram overlap.

5.6.4.5 ROUGE-L

ROUGE-L evaluates the similarity between the generated and reference responses using the Longest Common Subsequence (LCS).

The F1-based ROUGE-L score is calculated as:

$$ROUGE-L = \frac{2 \times P_L \times R_L}{P_L + R_L} \quad (5.18)$$

where P_L and R_L represent precision and recall calculated using the Longest Common Subsequence.

The system obtained:

$$\boxed{ROUGE-L = 0.305} \quad (5.19)$$

This indicates a moderate level of sequence-level similarity between the generated responses and the reference answers.

5.6.4.6 Token-level F1

Token-level F1 measures the overlap of tokens between the generated response and the reference answer while considering both precision and recall.

The metric is calculated as:

$$F1 = \frac{2 \times P \times R}{P + R} \quad (5.20)$$

where P represents token-level precision and R represents token-level recall.

The system achieved:

$$\boxed{\text{Token F1} = 0.330} \quad (5.21)$$

This indicates that approximately 33.0% of the normalized tokens overlap between the generated and reference answers.

5.6.5 Generation Evaluation Results

The generation results are presented in Table 5.2.

Table 5.2: Generation Evaluation Metrics

Metric	Average Score
Exact Match	0.100
BLEU	0.170
ROUGE-1	0.345
ROUGE-2	0.210
ROUGE-L	0.305
Token-level F1	0.330

5.6.6 Analysis of Generation Results

The generation results show different levels of lexical similarity between the generated responses and the reference answers. ROUGE-1 obtained the highest score of 0.345, followed by Token-level F1 with 0.330 and ROUGE-L with 0.305. ROUGE-2 achieved a score of 0.210, while BLEU obtained 0.170.

The Exact Match score of 0.100 is comparatively low. This is expected for open-ended interview responses because the generated response does not necessarily need to reproduce the reference answer word-for-word.

For example, the reference answer may describe a candidate’s strength using one particular sentence structure, while the generated answer may communicate the same idea using different words or additional explanation. Such responses can therefore have limited exact or n-gram overlap while still providing a relevant answer.

The results should therefore be interpreted primarily as measures of lexical and sequence overlap rather than as direct measures of the overall correctness or usefulness of the generated interview responses.

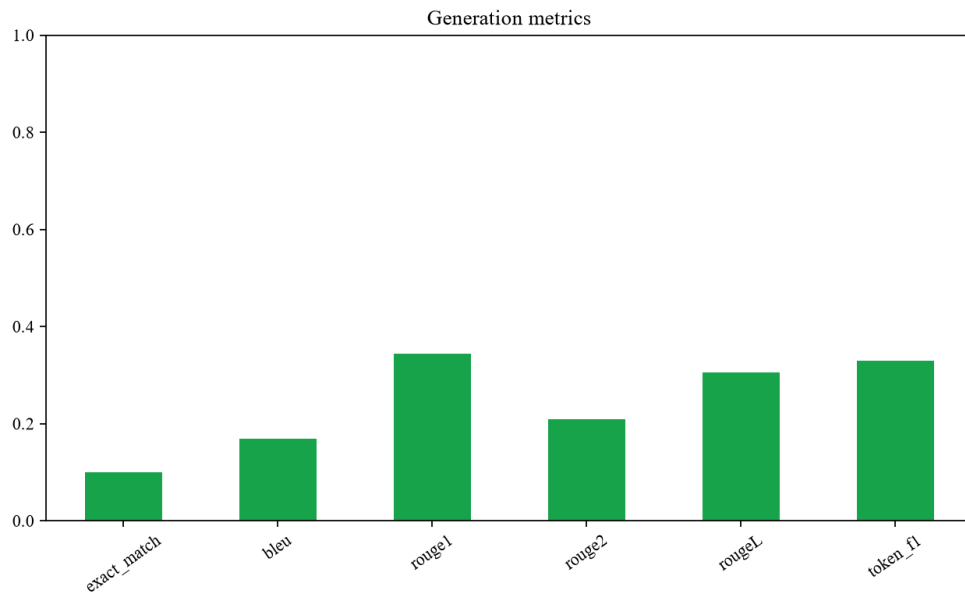


Figure 5.7: Generation Metrics Result

Chapter 6

Dataset Collection and Analysis

6.1 Data Collection and Data Description

To generate and evaluate behavioural interview questions across varying levels of difficulty, question–answer data was collected from multiple publicly available sources. Each source was chosen to represent a specific difficulty tier used by the system’s adaptive question generation module: simple, medium, and hard.

6.1.1 Data Sources

6.1.1.1 Easy-Level Questions

Easy-level questions focus on fundamental behavioural and situational competencies that assess a candidate’s basic professional knowledge, communication skills, and work ethic. These questions typically ask candidates to share straightforward experiences, demonstrate core competencies, and provide direct, accessible answers. The dataset comprises 320 easy-level questions sourced from a publicly available dataset on Kaggle titled “HR Interview Questions and Ideal Answers” [4], which is provided in JSON format and consists of commonly asked HR interview questions paired with corresponding ideal answers. This dataset was selected for its structured format and suitability as a reference for evaluating basic-level candidate responses.

Example Easy-Level Question: “Tell me about a time you had to learn something completely new quickly.”

The expected answer focuses on eagerness to learn, adaptability, and a concrete example of successfully acquiring new skills.

Source: https://www.kaggle.com/datasets/aryan208/hr-interview-questions-and-ideal-answers?select=hr_interview_questions_dataset.json

6.1.1.2 Medium-Level Questions

Medium-level questions assess deeper problem-solving abilities, strategic thinking, and role-specific expertise. These questions present realistic scenarios and ask candidates to demonstrate judgment in decision-making, analytical thinking, and understanding of industry practices. The dataset comprises 150 medium-level questions collected from Exponent [5], an interview-preparation platform that contains real interview questions shared by candidates across technical and behavioural interview rounds. Relevant behavioural and situational questions were manually filtered from this platform to align with the scope of the proposed system.

Example Medium-Level Question: “Tell me about a product you love. How would you improve it?”

The expected answer demonstrates product thinking, user empathy, critical analysis, and articulation of concrete improvements based on user pain points.

Source: <https://www.tryexponent.com/questions>

6.1.1.3 Hard-Level Questions

Hard-level questions present complex, multi-faceted scenarios that demand advanced technical knowledge, strategic decision-making, and leadership-oriented thinking. These questions require candidates to navigate ambiguous situations, consider multiple constraints, and provide nuanced solutions that demonstrate domain expertise. The dataset comprises 20,444 hard-level questions collected from the “InterviewForge_GenDS” dataset hosted on Hugging Face [6]. This dataset provides generated behavioural interview question–answer pairs suited for advanced, scenario-based, and leadership-oriented interview scenarios, making it appropriate for representing the hard difficulty tier in the proposed system.

Example Hard-Level Question: “You’ve been tasked with implementing a new network infrastructure for an IoT device farm currently consisting of 500 devices. What are the key requirements, constraints, and trade-offs you would consider?”

The expected answer demonstrates technical depth, systems thinking, risk assessment, and the ability to balance competing priorities in a complex infrastructure project.

Source: https://huggingface.co/datasets/Davichick/InterviewForge_GenDS/viewer/default/train

6.1.2 Data Format and Structure

The Kaggle dataset was provided in JSON format with predefined fields for question and answer text. The Exponent dataset, in contrast, was collected as unstructured web text and required manual extraction into a consistent question–answer format. The Hugging Face dataset was accessed through its dataset viewer in a tabular train-split format, with structured question and answer columns readily usable for direct integration.

6.1.3 Dataset Summary

The following table summarizes the data sources used.

Table 6.1: Summary of Data Sources by Difficulty Level

Difficulty Level	Source	Format
Simple	Kaggle (HR Interview Q&A Dataset)	JSON
Medium	Exponent (tryexponent.com/questions)	Web-curated text
Hard	Hugging Face (InterviewForge_GenDS)	Tabular (train split)

6.2 Data Preprocessing

After downloading, all three datasets were converted into a unified Microsoft Excel (.xlsx) format to standardize storage and simplify handling across the pipeline. The following preprocessing steps were then applied to prepare the data for use in the system:

1. **Format conversion:** The Kaggle JSON file, the manually curated Exponent question set, and the Hugging Face tabular dataset were each converted into individual .xlsx sheets to bring all sources into a single, consistent working format.
2. **Column filtering:** Unnecessary and redundant columns present in the original datasets, such as metadata fields not relevant to question generation or evaluation, were removed, retaining only the essential fields required by the system.
3. **Data cleaning:** Inconsistent formatting, missing values, and malformed entries were corrected across all sheets to ensure uniformity.
4. **Duplicate removal:** Duplicate question–answer pairs, both within and across the three sources, were identified and removed to avoid redundancy and bias during question generation.
5. **Difficulty labeling:** Questions that were not already labeled with a difficulty level were manually tagged as simple, medium, or hard based on their source and content.
6. **Embedding generation:** The cleaned and structured dataset was converted into dense vector embeddings using the *nomic-embed-text* model served locally through Ollama. Each question and its corresponding reference answer were embedded to capture their semantic meaning, enabling similarity-based retrieval rather than exact keyword matching during mock interview sessions.

The resulting cleaned, de-duplicated, and embedded dataset served as the unified input for the system’s question generation and response evaluation modules.

6.2.1 Processed Dataset Summary

Table 6.2 summarizes the data sources, their original formats, and the final processed state used in the system.

Table 6.2: Summary of Data Sources by Difficulty Level

Difficulty Level	Source	Original Format	Processed Format
Simple	Kaggle (HR Interview Q&A Dataset)	JSON	XLSX (cleaned, embedded)
Medium	Exponent (tryexponent.com/questions)	Web-curated text	XLSX (cleaned, embedded)
Hard	Hugging Face (Interview-Forge_GenDS)	Tabular (train split)	XLSX (cleaned, embedded)

6.3 Exploratory Data Analysis (EDA)

After preprocessing and embedding generation, a comprehensive exploratory data analysis (EDA) was conducted to understand the characteristics, distribution, and patterns of the compiled dataset across difficulty levels and job roles.

6.3.1 Easy-Level Dataset Analysis

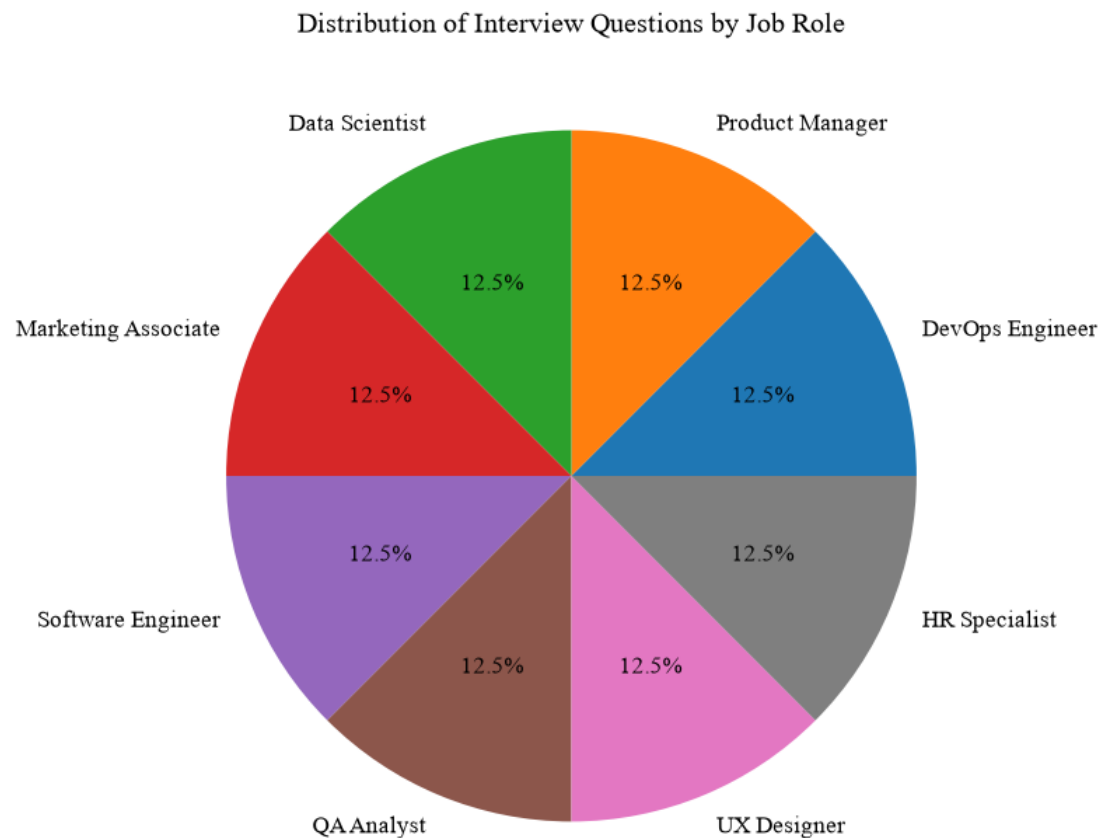


Figure 6.1: Bar Chart for Easy Questions per Role

The distribution of interview questions across 8 primary job roles (Data Scientist, Product Manager, DevOps Engineer, HR Specialist, UX Designer, QA Analyst, Software Engineer, and Marketing Associate) shows uniform 12.5% representation per role, ensuring balanced coverage for fundamental competencies.

6.3.2 Medium-Level Dataset Analysis

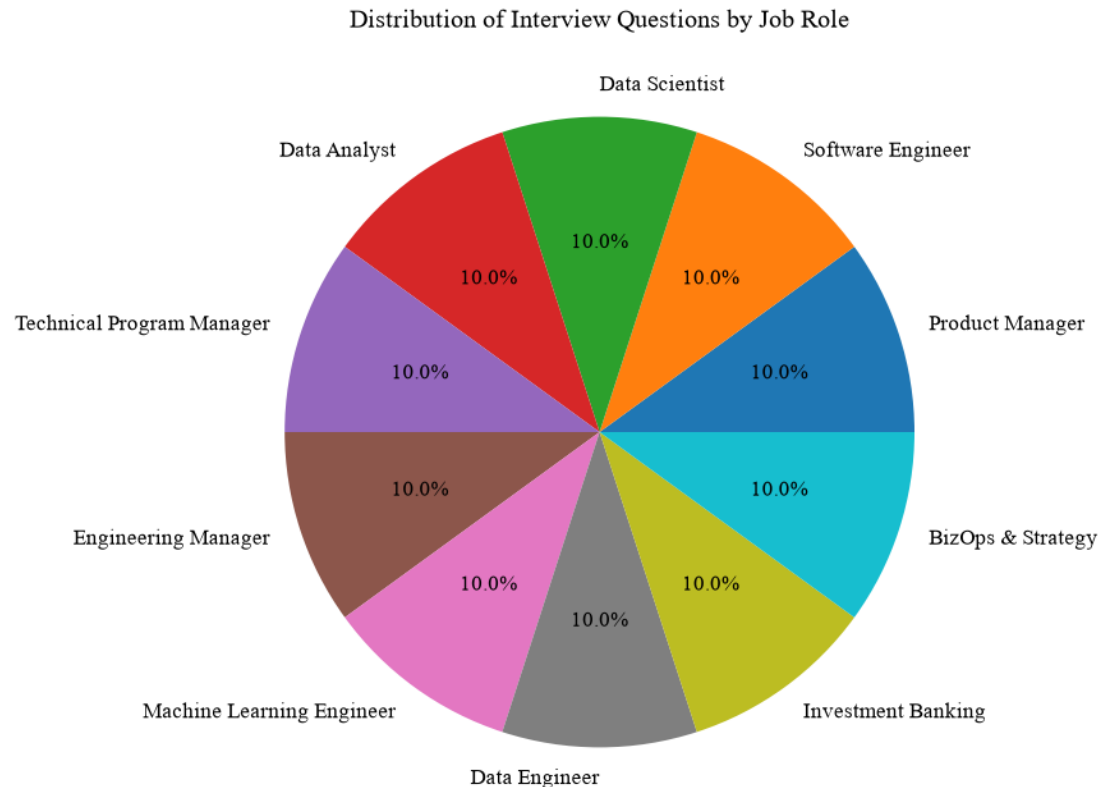


Figure 6.2: Bar Chart for Medium Questions per Role

Distribution across 10 specialized IT roles (Data Scientist, Software Engineer, Product Manager, DevOps Engineer, BizOps & Strategy, Investment Banking, Data Engineer, Machine Learning Engineer, Data Analyst, and Technical Program Manager) shows consistent 10% per-role allocation, representing intermediate career positions requiring situational judgment.

6.3.3 Hard-Level Dataset Analysis

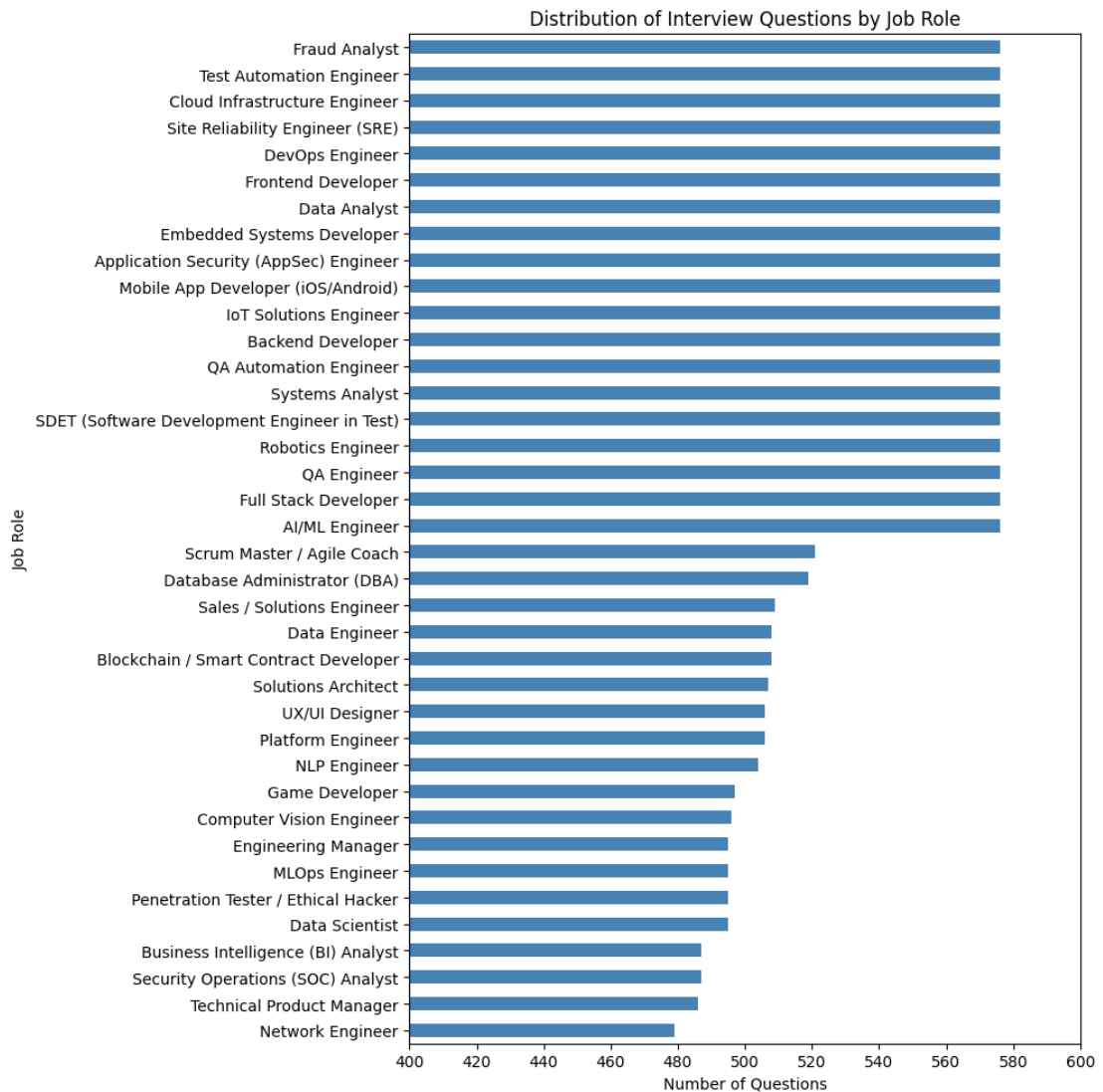


Figure 6.3: Bar Chart for Hard Questions per Role

Distribution across 40+ specialized and senior-level job roles including Fraud Analyst, Test Automation Engineer, Cloud Infrastructure Engineer, Site Reliability Engineer, DevOps Engineer, and numerous domain-specific positions shows varying bar lengths reflecting role-specific question volumes.

6.3.4 Comparative Analysis Across All Difficulty Levels

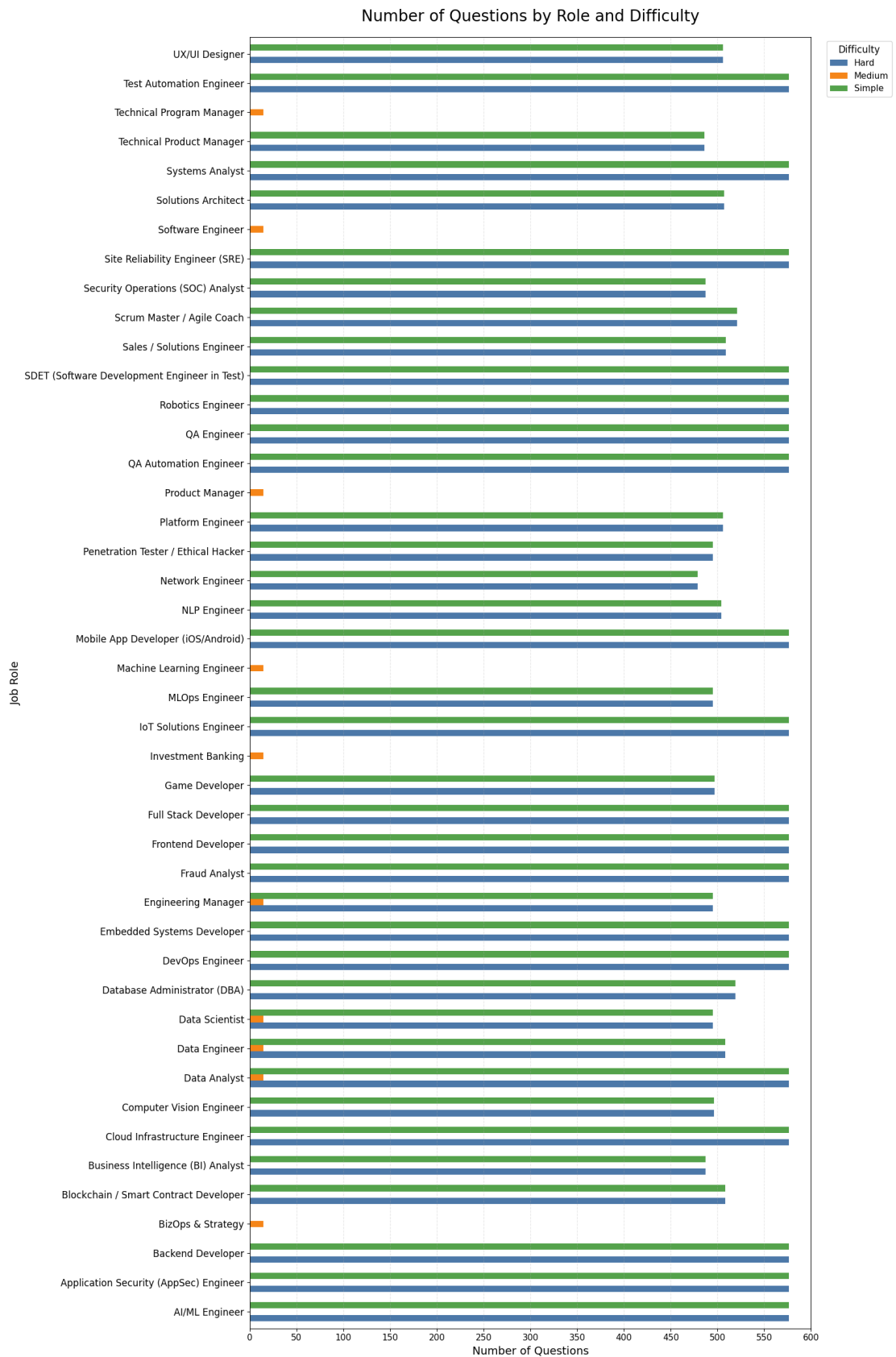


Figure 6.4: Comparative Analysis Across All Difficulty Levels

The grouped bar chart displays the number of questions per job role, clustered by difficulty tier (Hard, Medium, Simple). Most roles show comparable Hard and Simple question counts, typically in the 480–580 range, reflecting balanced coverage at these two tiers. In contrast, only a small subset of roles (e.g., Data Scientist, Product Manager, Software Engineer, Engineering Manager) contain Medium-tier questions, each with markedly fewer counts (roughly 10–20), highlighting the comparatively limited size of the Medium dataset relative to the Hard and Simple tiers.

6.3.5 Dataset Coverage Summary

The aggregated analysis highlights the following key characteristics:

- **Role Diversity:** Covers 40+ IT job roles, with 100–500+ questions per role.
- **Difficulty Stratification:** Questions are organized into simple, medium, and hard levels to support adaptive difficulty.
- **Answer Complexity:** Question complexity increases with difficulty, requiring progressively more detailed answers.
- **Balanced Distribution:** Each role includes questions across foundational, intermediate, and advanced skill levels.
- **Semantic Embedding:** Questions and reference answers are embedded using `nomic-embed-text` for semantic similarity-based evaluation.

This comprehensive dataset architecture provides the foundation for the system’s adaptive questioning mechanism and personalized evaluation methodology described in subsequent sections.

Chapter 7

Results And Discussion

7.1 Work Done

7.1.1 System Development and Interview Question Dataset

The proposed system, Generative AI-Based Behavioural Interview Preparation and Guidance System, has been developed to assist users in preparing for behavioural interviews. The system uses a question dataset containing interview questions associated with different job roles and difficulty levels. The dataset provides the knowledge base from which relevant interview questions can be retrieved and used during the interview preparation process. The questions are organized according to the available roles and difficulty categories, allowing the system to provide more relevant interview content to the user.

7.1.2 Question Retrieval Process

The retrieval component is responsible for finding questions that are semantically related to the user's input. When a user provides an interview-related question or input, the system processes the input and compares it with the questions stored in the vector database. The vector retrieval process calculates the distance between the user's input and stored questions. This distance is converted into a similarity score using:

Similarity Score = $1 - \text{Distance}$

The system retrieves the top five most similar questions for each input. These retrieved questions represent the candidates that can be used as relevant context for the subsequent response generation process.

7.1.3 Retrieval Evaluation and Analysis

The retrieval component was evaluated using Similarity Score, Precision@5, Recall@5, Mean Reciprocal Rank (MRR), and F1-score. For each evaluation query, the top five retrieved questions were compared with the relevant questions according to the evaluation criterion.

The average Similarity Score of 0.685 indicates that the retrieved questions have a moderate to relatively high semantic similarity to the input questions. Precision@5 achieved 0.252, meaning that approximately 25.2

The system achieved a high MRR of 0.958, which indicates that the first relevant question is generally positioned very close to the top of the retrieved results. Finally, the retrieval F1-score of 0.295 represents the balance between Precision@5 and Recall@5. These results show that although the system has room for improvement in retrieving a larger

proportion of relevant questions, it is generally successful at placing a relevant result near the top of the retrieval list.

7.1.4 Context-Based Response Generation

After retrieving relevant interview questions, the retrieved information can be used as context for the generative component of the system. Instead of relying only on the user's input, the generation process uses the retrieved interview-related information to produce a response appropriate to the given query. This retrieval-and-generation approach helps connect the user's question with the available interview knowledge in the dataset.

The generated responses can then be evaluated using Exact Match, BLEU, ROUGE-1, ROUGE-2, ROUGE-L, and Token-level F1. These metrics measure different aspects of the generated response, including exact agreement, word overlap, sequence similarity, and token-level similarity with the expected answer.

7.1.5 Mock Interview Functionality

The system provides a mock interview functionality through which users can practice behavioural interview questions. During the mock interview, questions are presented to the user, and the user submits answers based on their own experience and understanding. The mock interview process provides the foundation for evaluating the user's responses and generating feedback. The retrieval component supports this process by providing relevant interview questions based on the available dataset.

7.2 Web Interface

7.2.1 Home Page

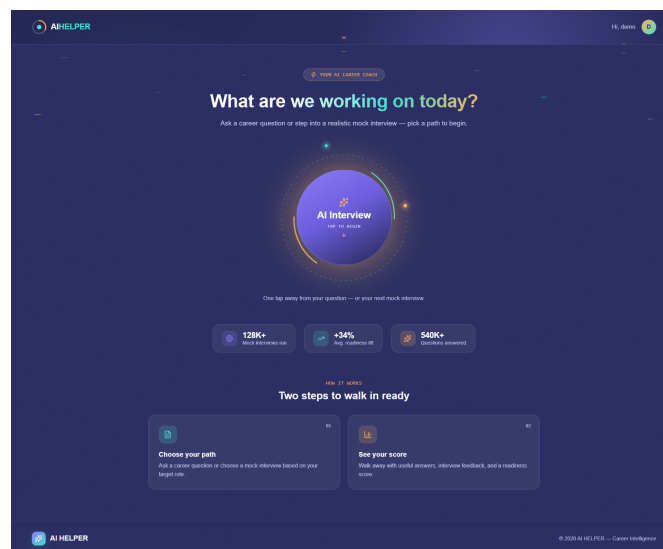


Figure 7.1: Home Page

The primary dashboard serves as the central entry point for the platform, offering two main layout variations to guide the user journey. Image 1 features a centralized hero

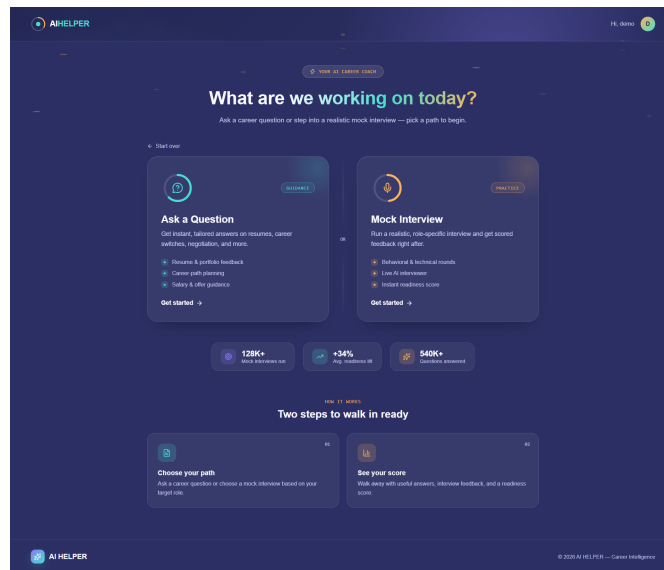


Figure 7.2: Option Page

call-to-action (“AI Interview - Tap to Begin”) flanked by key platform performance metrics like total mock interviews run and questions answered. Image 2 splits this hero area into two distinct choices—“Ask a Question” for general career guidance and “Mock Interview” for structured practice. Both variations anchor the bottom section with a clear two-step onboarding process, establishing immediate clarity and social proof.

7.2.2 Account Creation

Create your account
Start your journey toward interview success.

Full Name
Enter your full name

Email Address
you@example.com

Password
Create a password

Confirm Password
Confirm your password

☐ I agree to the Terms of Service and Privacy Policy.

Create Account

Already have an account? [Login](#)

Welcome back
Sign in to continue your interview preparation.

Email Address
you@example.com

Password
Enter your password

[Forgot password?](#)

☐ Remember me

Login

OR

Continue with Google

Don't have an account? [Create account](#)

(a) Register

(b) Login

Figure 7.3: Account Creation Interface

The registration and login interfaces utilize a cohesive dark purple aesthetic featuring rounded form cards to handle user access. Image 3 provides a standard multi-field account creation workflow requiring full name, email, password confirmation, and terms agreement. Image 4 simplifies returning user access with standard credentials, a

”Remember me” toggle, and a single-click ”Continue with Google” social login option. Both screens emphasize user experience through visible password toggles and clear call-to-action buttons styled in bright gradient fills.

7.2.3 AI Chat

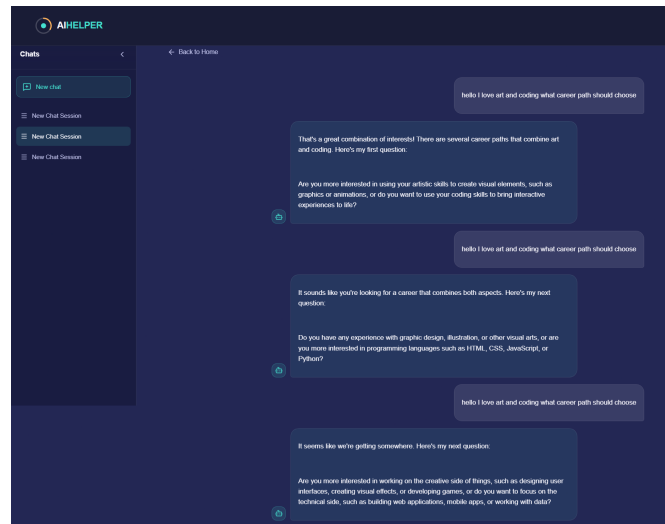
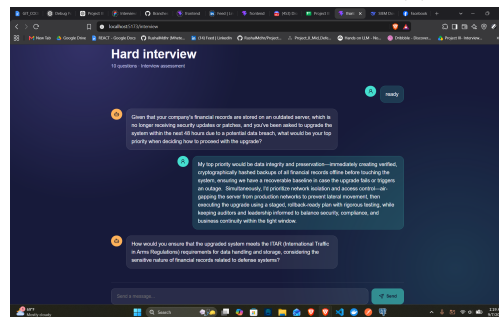


Figure 7.4: AI Chat Interface

A dedicated career advice interface with a collapsible sidebar listing active chat sessions. The dialogue shows a user asking for career recommendations based on interests in art and coding. The AI system guides the conversation using targeted follow-up questions to narrow down specific skill sets like UI design, game development, or visual web development.

(b) Medium Mock Interview



(c) Hard Mock Interview

These three screens illustrate the conversational AI engine adapting across different engagement types and technical depths. Image 6 (Aichat.png) showcases a open-ended career advisor interface where the AI asks targeted follow-up questions to help a user combine art and coding interests into a career path. Image 7 (easy.jpg) captures an entry-level technical interview centered on past projects, embedding architectures, and team collaboration dynamics. Image 8 (8.jpg) demonstrates a high-stakes, scenario-based technical assessment where the candidate answers complex questions regarding database integrity, emergency rollbacks, and ITAR compliance.

7.3 Remaining Work

7.3.1 User Profile and Dashboard

The user profile and dashboard remain to be completed. The profile component should allow users to maintain their personal and interview-related information, including relevant skills, selected job role, and other information required for interview preparation. This information can be used to provide a more organized and personalized experience within the system.

7.3.2 User Interview Evaluation

The interview evaluation component is one of the major remaining functionalities. After the user completes a mock interview and submits their answers, the system should evaluate the responses based on the expected answer or relevant reference information. The evaluation should determine how closely the user's response matches the expected content while considering the meaning and important information contained in the response.

The evaluation process should use the selected generation metrics, including Exact Match, BLEU, ROUGE-1, ROUGE-2, ROUGE-L, and Token-level F1, where applicable. These metrics provide different measurements of the similarity between the user's response and the reference answer. The resulting scores should be combined with the interview response analysis to provide understandable feedback to the user.

7.3.3 Interview Score and Feedback Presentation

After evaluating the submitted answers, the system should present the results in a clear format. The results page should show the user's performance for individual questions as well as the overall interview performance. Appropriate feedback should explain areas where the response was relevant and areas where improvement may be required.

The purpose of this component is not only to provide a numerical score but also to help users understand their performance and improve their behavioural interview responses.

7.3.4 Interview History

A further remaining component is the storage and presentation of previous interview results. Completed mock interviews should be associated with the user's profile so that the user can review previous attempts. The system can present previous scores and feedback to help users monitor their progress across multiple practice sessions.

Bibliography

- [1] S. Kulkarni, M. Madane, D. Garule, and A. Kore, “PrepWise: A generative AI-powered personalized interview preparation and multi-dimensional assessment platform,” *International Journal of Electrical Engineering and Ethics*, vol. 9, no. 2, pp. 1347–1352, mar 2026. [Online]. Available: <https://ijetjournal.org/wp-content/uploads/PrepWise-A-Generative-AI-Powered-Personalized-Interview-Preparation-and-Multi-Dimensional-Assessment-Platform.pdf>
- [2] K. H. Gharte, N. Pawar, and A. Bijwe, “AI-powered career recommendation system using hybrid architecture, semantic retrieval, and dynamic scoring,” *International Journal on Advanced Computer Theory and Engineering*, vol. 15, no. 2S, pp. 91–97, 2026. [Online]. Available: <https://journals.mriindia.com/index.php/ijacte/article/view/2976>
- [3] “SimInterview: Explainable AI interview assistant using LLMs and RAG,” PubMed Central, 2025. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11564924/>
- [4] Aryan, “HR interview questions and ideal answers dataset,” Kaggle, 2026, accessed: 2026-09-05. [Online]. Available: https://www.kaggle.com/datasets/aryan208/hr-interview-questions-and-ideal-answers?select=hr_interview_questions_dataset.json
- [5] “Exponent interview questions,” Exponent, 2026, accessed: 2026-09-05. [Online]. Available: <https://www.tryexponent.com/questions>
- [6] Davichick, “InterviewForge_GenDS dataset,” Hugging Face, 2026, accessed: 2026-09-05. [Online]. Available: https://huggingface.co/datasets/Davichick/InterviewForge_GenDS/viewer/default/train